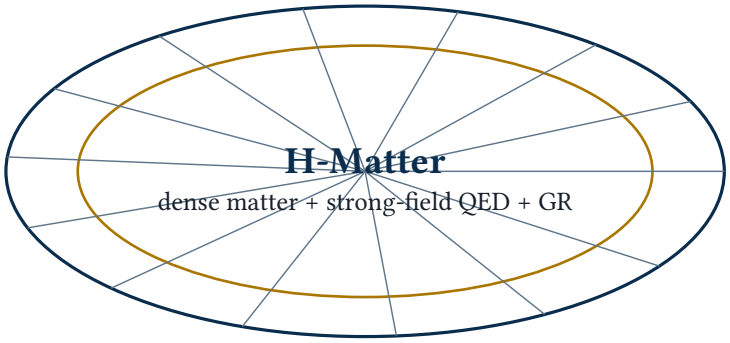
HMGQM Theory

Hyper-Dense Magneto-Gravitational Quantum Matter

Complete Revised Formulation: From Original Hypothesis to a Self-Consistent Effective Framework



axisymmetric, horizon-free, testable regime

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Version 3.0 – Complete Formulation

October 2026

Scientific status: exploratory theoretical framework. This document is deliberately written so that every strong claim is separated from the underlying established physics. It does not assert experimental verification of H-Matter, quantum gravity, or a new fundamental phase of matter.

Document Status and Reader Guide

This volume consolidates the author's original HMGQM hypothesis and the subsequent self-consistency revision into a single, substantially expanded formulation. The original draft proposed H-Matter as a previously undefined anisotropic phase emerging when extreme gravity, ultra-strong magnetic fields, and QED vacuum effects act together. The second version narrowed that claim and replaced the inconsistent parts with a classical-GR, strong-field-QED, dense-matter framework. The present version develops that repaired structure from first principles at the level appropriate to an effective theory, and adds explicit derivations, limiting cases, numerical requirements, observational discipline, and a falsifiability program.

The central change is conceptual as much as mathematical. The document no longer assumes that anisotropy automatically creates a new fundamental substance, nor that a magnetized compact object can be treated by a spherical anisotropic TOV equation after axial symmetry has been invoked. Instead, HMGQM is formulated as a coupled problem involving a relativistic dense-matter grand potential, Landau quantization of charged species, a renormalized one-loop Heisenberg-Euler sector for QED vacuum response, the electromagnetic field stress tensor, and the stationary axisymmetric Einstein-Maxwell equations. A genuinely new phase is reserved for the case in which a thermodynamic nonanalyticity or a separately derived order parameter is demonstrated.

How to read the theory. The words *established*, *model assumption*, *derived within the model*, and *prediction* are used deliberately. Established physics supplies the mathematical ingredients. Model assumptions specify where the approximation is used. Derived quantities follow from those assumptions. A prediction is a falsifiable quantity produced only after the full equilibrium and perturbation problems are solved.

Abstract

We formulate HMGQM as a high-field, high-density effective framework for compact objects in which magnetic-field geometry, Landau quantization, QED vacuum polarization, and relativistic gravity are simultaneously important. The minimal theory uses classical general relativity coupled to a renormalized one-loop Heisenberg-Euler electromagnetic sector and an explicit microscopic dense-matter model. In a local comoving frame, charged fermions are quantized into Landau levels, while neutral species and anomalous magnetic moments are treated separately. The thermodynamic grand potential determines magnetization and the principal stresses, avoiding double counting of electromagnetic and vacuum contributions. Global equilibrium is posed as a stationary axisymmetric Einstein-Maxwell-matter boundary-value problem with an exterior electrovacuum or magnetospheric match. Stability is treated hierarchically: local thermodynamic consistency, global equilibrium sequences, and finally linear dynamical perturbations.

The resulting framework defines an operational H-regime but does not presuppose a new phase transition. The principal observational channels are polarization-dependent photon propagation in magnetized curved spacetime, magnetic deformation and ellipticity, changes in mass-radius and tidal response, and magnetic splitting of oscillation spectra. Existing observations, including GW170817, GW190814, NICER radius measurements, and recent magnetar polarization measurements, are treated as constraints or baselines rather than as evidence that the proposed H-Matter phase already exists. The final section states the exact conditions under which the framework would be falsified or elevated from a regime label to a physically distinct phase.

Keywords

Heisenberg-Euler; **magnetic anisotropy**; **Einstein-Maxwell**; **chiral EFT**; **vacuum birefringence**; **GW170817**; **GW190814**; **NICER**; **IXPE**.

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1 From the Original Hypothesis to the Complete Formulation

1.1 The original physical intuition

The original HMGQM draft began with an intuitive question: what happens when a compact star simultaneously reaches extremely high density, extremely strong magnetic field, and sufficiently strong relativistic gravity that the usual picture of nearly isotropic nuclear matter becomes inadequate? The draft proposed that classical atomic structure is destroyed, charged particles become magnetically organized, vacuum polarization becomes dynamically relevant, and the resulting stress ceases to be represented by a single scalar pressure. It then named the putative regime H-Matter. The core motivation remains valuable because each ingredient corresponds to a real physical effect: degenerate fermionic matter at supranuclear density, Landau quantization of charged particles, electromagnetic field stress, nonlinear QED response, and general-relativistic self-gravity.

The original document also made stronger claims than its equations could support. It mixed Planck-scale language with a classical spacetime assumption, used a finite weak-field Euler-Heisenberg expansion deep into the strong-field regime, represented charged matter by a three-dimensional Fermi integral while simultaneously invoking Landau collapse, used a spherical anisotropic TOV equation after claiming axial symmetry, and treated GW170817 and GW190814 as if they supplied direct post-merger frequency points. These mismatches motivated the complete reformulation presented here.

1.2 The v2.0 repair, now developed fully

The second version converted the idea into an effective framework. It made classical GR explicit, replaced the truncated QED sector with the finite-field one-loop Heisenberg-Euler proper-time form, introduced Landau-level sums, derived the thermodynamic principal stresses from a single grand potential, and posed the global problem in stationary axisymmetric form. It also reclassified H-Matter as a regime label until a genuine phase transition can be shown. Those repairs are preserved here, but the present version goes beyond a repair list: it supplies the derivation chain and states the exact mathematical closure required for a numerical implementation.

Central thesis of HMGQM v3.0. A compact object can enter an operational high-field/high-density regime in which the field-induced microscopic quantization of charged species and the electromagnetic/QED contributions to stress-energy are not perturbative relative to the dense-matter stress. In that regime, the physically correct global description is axisymmetric and coupled. Whether this regime constitutes a new thermodynamic phase is a separate question that must be answered by the microscopic thermodynamics.

1.3 Established ingredients versus proposed synthesis

The theoretical ingredients are not claimed as discoveries. Landau quantization is standard relativistic quantum mechanics. The Heisenberg-Euler effective action is a standard QED result. Einstein-Maxwell theory is a standard classical relativistic field framework. Magnetized neutron-star equilibrium has already been studied numerically in self-consistent axisymmetric calculations. The proposed contribution is the specific synthesis: define an operational H-regime, use a thermodynamically consistent strong-field EOS, couple it to an axisymmetric relativistic star, and ask whether that combined system produces a qualitatively distinct stable branch or a robust, discriminating observational signature.

2 Physical Foundations

2.1 Ultra-dense matter

At densities above nuclear saturation, matter is no longer adequately described by an ideal gas of nucleons. The energy per particle depends on strong interactions, composition, many-body correlations, and the symmetry energy. A quantitative model therefore requires a specified microscopic closure. Candidates include relativistic mean-field models, chiral effective field theory at densities where it is controlled, phenomenological calibrated equations of state, hyperonic models, or quark-matter descriptions. HMGQM does not invent a universal effective mass function and does not claim that one composition remains valid at all densities.

The baryon number density is denoted by n_B. Nuclear saturation is used only as a reference scale,

$$n_{\rm sat} \simeq 0.16\ \mathrm{fm}^{-3}, \tag{1}$$

corresponding to a mass density of order $\rho_{\rm sat} \simeq 2.8 \times 10^{17}\ \mathrm{kg\,m^{-3}}$. The notation $n_B \gtrsim n_{\rm sat}$ therefore means *supranuclear*, not automatically exotic.

2.2 Strong magnetic fields

A magnetic field introduces a preferred spatial direction. For charged particles, the transverse momentum becomes discretized and the continuous three-dimensional phase-space measure is replaced by a sum over Landau levels. Even before QED vacuum corrections become important, the field can therefore alter the matter equation of state and the shape of a compact star. Once the classical field energy density $B^2/2$ becomes a substantial fraction of the matter pressure or energy density, electromagnetic stress must be treated as part of the global equilibrium rather than as a small perturbation.

A particularly useful dimensionless diagnostic is

$$\beta_B \equiv \frac{B^2/2}{P_m}, \tag{2}$$

where P_m is the matter-sector thermodynamic pressure in natural units. The H-regime does not require $\beta_B \sim 1$ everywhere; it requires only that the magnetic contribution not be safely negligible for the quantity being computed.

2.3 Quantum electrodynamic vacuum response

The electron mass sets the QED critical field

$$B_{\rm QED} = \frac{m_e^2}{e} \simeq 4.414 \times 10^9\ \mathrm{T} \simeq 4.414 \times 10^{13}\ \mathrm{G}. \tag{3}$$

For $B \gtrsim B_{\rm QED}$, the QED vacuum becomes a nonlinear optical medium. For magnetars and other ultra-magnetized objects this can affect photon polarization transport. Importantly, the same QED sector also contributes to the electromagnetic effective action and therefore to the stress tensor. HMGQM treats these effects as a single action-derived sector rather than adding a separate "vacuum pressure" by hand.

2.4 Curved spacetime

The minimal version of HMGQM retains classical general relativity. The local microphysics is evaluated in a local inertial frame, while gravity couples to the total stress-energy and determines the global equilibrium geometry. The theory therefore does not assume that ordinary stellar curvature is itself a quantum-gravity regime. A quantum-gravitational extension would require new degrees of freedom

and a separate action; it is outside the minimal framework.

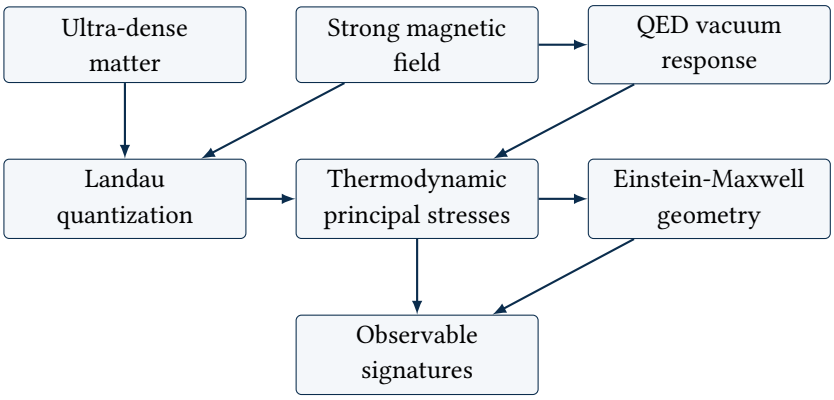


Figure 1: The physical chain retained in the complete formulation. The novelty is not any individual block; it is the self-consistent coupling of the blocks in one equilibrium and perturbation problem.

3 Conventions and Domain of Validity

3.1 Natural units and covariant notation

Unless a dimensional quantity is explicitly quoted in SI or cgs units, the equations use natural Heaviside-Lorentz units $\hbar = c = k_B = \epsilon_0 = \mu_0 = 1$ and metric signature $(-,+,+,+)$. The electromagnetic field tensor is $F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$, and its dual is $\star F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}$.

The electromagnetic invariants are

$$\mathscr{F} = \frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad \mathscr{G} = \frac{1}{4}F_{\mu\nu}\star F^{\mu\nu}. \tag{4}$$

For a purely magnetic field measured by a comoving observer, $\mathscr{F} = B^2/2$ and $\mathscr{G} = 0$.

3.2 Operational H-regime diagnostics

No single numerical field threshold is sufficient to define H-Matter. Instead we define a set of dimensionless diagnostics:

$$\chi_B \equiv \frac{B}{B_{\rm QED}}, \tag{5}$$

$$\nu_{{\rm max},i} = \left\lfloor \frac{\mu_i^{*2} - m_i^{*2}}{2|q_i|B} \right\rfloor, \tag{6}$$

$$\beta_B \equiv \frac{B^2/2}{P_m}, \tag{7}$$

$$\kappa_{\rm loop} \equiv \alpha \ln(1+\chi_B), \tag{8}$$

$$\chi_R \equiv \left(\frac{\lambda_e}{L_{\rm curv}}\right)^2, \tag{9}$$

where $\lambda_e = 1/m_e$ is the electron Compton wavelength and $L_{\rm curv}$ is a characteristic local curvature radius.

We define the H-regime operationally by

$$\chi_B \gg 1, \qquad n_B \gtrsim n_{\rm sat}, \qquad \nu_{{\rm max},i} = O(1) \text{ for at least one dynamically important charged species}, \qquad \beta_B \ll 1, \tag{10}$$

with the additional consistency requirements

$$\kappa_{\rm loop} \ll 1 \quad \text{(for one-loop control)}, \qquad \chi_R \ll 1 \quad \text{(for the minimal local-QED approximation)}. \tag{11}$$

These are diagnostics, not phase boundaries.

Important domain statement. The H-regime is not "the Planck regime." It is a sub-Planckian, classical-GR effective-theory regime. If curvature approaches a genuinely quantum-gravitational scale, the present theory must be declared out of domain rather than extrapolated.

3.3 Field-variation and photon-energy constraints

The local constant-field approximation underlying the simplest Heisenberg-Euler description requires the background electromagnetic field to vary slowly on the electron Compton scale and the probe photon to remain in a low-energy regime relative to pair-creation thresholds. In practical notation one needs conditions of the schematic form

$$\lambda_e |\nabla B|/B \ll 1, \qquad \omega_\gamma / m_e \ll 1, \tag{12}$$

with the exact validity limits assessed for the intended calculation. When these conditions fail, derivative corrections or a more complete strong-field QED calculation is required.

4 Fundamental Effective Action

4.1 Action principle

The minimal complete HMGQM action is

$$S = \int {\rm d}^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \Delta \mathscr{L}_{\rm HE}(F, \star F) + \mathscr{L}_{\rm matter}(\Psi, A, g; \theta_{\rm micro}) \right]. \tag{13}$$

Here R is the Ricci scalar, $\Delta\mathscr{L}_{\rm HE}$ is the renormalized one-loop Heisenberg-Euler correction, and $\mathscr{L}_{\rm matter}$ contains the chosen microscopic dense-matter model. The parameter vector $\theta_{\rm micro}$ collects whichever strong-interaction couplings, effective masses, meson fields, bag constants, or tabulated EOS parameters the numerical implementation uses.

This action is the central closure of the theory. It fixes the bookkeeping convention: all non-gravitational sources are obtained by varying one action. No separate "magnetic pressure," "vacuum pressure," and "matter pressure" may be added again after the total tensor has already been derived.

4.2 Metric variation

Varying the non-gravitational part with respect to the metric gives

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_{\rm non-grav}}{\delta g^{\mu\nu}}, \tag{14}$$

and therefore

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{15}$$

Diffeomorphism invariance implies

$$\nabla_\mu T^{\mu\nu} = 0. \tag{16}$$

Equation (16) is a consistency identity of the coupled solution, not an optional condition.

4.3 Electromagnetic variation

Define the constitutive tensor

$$H^{\mu\nu} \equiv -2\frac{\partial(\mathscr{L}_{\rm EM} + \Delta\mathscr{L}_{\rm HE})}{\partial F_{\mu\nu}}, \tag{17}$$

where $\mathscr{L}_{\rm EM} = -\frac{1}{4}F^2$. Variation with respect to A_μ gives

$$\nabla_\mu H^{\mu\nu} = J^\nu, \qquad \nabla_{[\alpha}F_{\beta\gamma]} = 0. \tag{18}$$

The second relation is the Bianchi identity. The first contains the material current and the nonlinear QED constitutive response.

4.4 Why the action matters

In the original draft, vacuum corrections appeared in the pressure formula and again inside the stress tensor. The action formulation removes the ambiguity. Once $\Delta\mathscr{L}_{\rm HE}$ is specified and $\mathscr{L}_{\rm matter}$ is fixed, there is exactly one $T_{\mu\nu}$. Thermodynamic formulas are then interpreted as a local representation of that same tensor in the comoving frame.

5 Strong-Field QED: Renormalized Heisenberg-Euler Sector

5.1 Finite-field one-loop response

For a slowly varying, locally constant magnetic background, the renormalized one-loop spinor-QED correction for a pure magnetic field can be written in proper-time form as

$$\Delta\mathscr{L}_{\rm HE}(B) = -\frac{1}{8\pi^2}\int_0^\infty \frac{\mathrm{d}s}{s^3}e^{-m_e^2 s}\left[eBs\coth(eBs) - 1 - \frac{(eBs)^2}{3}\right]. \tag{19}$$

The subtraction terms implement vacuum-energy normalization and charge renormalization in the chosen convention. The defining feature is the non-polynomial dependence on B. The finite-field expression is therefore not replaced by a weak-field Taylor series in the H-regime.

5.2 Weak-field limit as a test, not a definition

For $B \ll B_{\rm QED}$, the proper-time integral expands as a series beginning schematically with

$$\Delta\mathscr{L}_{\rm HE} \propto \frac{\alpha^2}{m_e^4}\left[(F_{\mu\nu}F^{\mu\nu})^2 + \frac{7}{4}(F_{\mu\nu}\star F^{\mu\nu})^2\right] + \mathit{exthigherorders}. \tag{20}$$

This expansion is valuable because it must reproduce the familiar weak-field QED result. It is not the defining expression when $B \gg B_{\rm QED}$.

5.3 Magnetization and QED contribution

The QED sector modifies the constitutive relation through

$$M_{\mathrm{HE}} = -\frac{\partial \Delta\Omega_{\mathrm{HE}}}{\partial B}, \tag{21}$$

where the precise thermodynamic sign and identification of the effective potential follow the convention used in the matter-plus-vacuum grand potential. The total magnetization is obtained from the full renormalized grand potential.

5.4 One-loop control

The one-loop approximation is itself an effective truncation. A useful bookkeeping measure is $\kappa_{\mathrm{loop}} = \alpha \ln\!\left(1 + B/B_{\mathrm{QED}}\right)$. If this parameter ceases to be small, higher-loop corrections may no longer be negligible. The theory therefore contains its own control criterion: the H-regime can be physically interesting without being outside the perturbative hierarchy of the QED loop expansion.

> **Repair principle.** In v1, a finite F^4 expression was used while claiming a non-perturbative strong-field regime. In the complete formulation, the proper-time Heisenberg-Euler sector is the defining one-loop expression, and the weak-field F^4 expansion is used only as a limiting check.

5.5 Curved-spacetime caveat

Equation (19) is a local flat-background effective action. Curvature-dependent QED corrections exist in a broader effective field theory, but they require additional terms involving curvature invariants and field tensors. HMGQM v3.0 deliberately excludes those terms from the minimal model and requires $\chi_R \ll 1$ for the local treatment. If a future application violates that condition, the theory must be upgraded rather than silently extrapolated.

6 Microscopic Dense Matter

6.1 Minimal particle content

The minimal cold matter composition is $npe\mu$: neutrons, protons, electrons, and muons when energetically allowed. Hyperons, resonances, or quarks may appear at higher density, but their inclusion must be tied to a specific microscopic EOS. Charge neutrality in $npe\mu$ matter is

$$n_p = n_e + n_\mu, \tag{22}$$

and beta equilibrium is

$$\mu_n - \mu_p = \mu_e = \mu_\mu. \tag{23}$$

These relations generalize in the obvious way when additional species are present.

6.2 Landau spectrum of charged fermions

For a charged fermion i with charge q_i and effective mass m_i^* in a uniform local magnetic field, the minimally coupled relativistic spectrum is

$$E_{i\nu}(p_z) = \sqrt{p_z^2 + m_i^{*2} + 2\nu|q_i|B}, \qquad \nu = 0, 1, 2, \ldots \tag{24}$$

with spin degeneracy

$$g_\nu = 2 - \delta_{\nu 0}. \tag{25}$$

At zero temperature the Fermi momentum in level ν is

$$k_{F,i\nu} = \sqrt{\mu_i^{*2} - m_i^{*2} - 2\nu|q_i|B}, \tag{26}$$

whenever the radicand is non-negative. Thus

$$\nu_{\max,i} = \left\lfloor \frac{\mu_i^{*2} - m_i^{*2}}{2|q_i|B} \right\rfloor. \tag{27}$$

6.3 Number density

The density of a charged species is

$$n_i = \frac{|q_i|B}{2\pi^2} \sum_{\nu=0}^{\nu_{\max,i}} g_\nu k_{F,i\nu}. \tag{28}$$

This is the precise replacement for the ordinary three-dimensional momentum integral when Landau structure is dynamically important.

6.4 Grand potential

For a free relativistic charged species at $T = 0$, the matter contribution is

$$\Omega_i = -\frac{|q_i|B}{4\pi^2} \sum_{\nu=0}^{\nu_{\max,i}} g_\nu \left[\mu_i^* k_{F,i\nu} - m_{i\nu}^2 \ln\left(\frac{\mu_i^* + k_{F,i\nu}}{m_{i\nu}} \right) \right], \tag{29}$$

where

$$m_{i\nu}^2 = m_i^{*2} + 2\nu|q_i|B. \tag{30}$$

In an interacting model, this expression is replaced or supplemented by interaction terms consistent with the chosen closure. The HMGQM architecture remains the same: the full thermodynamic potential is a function of chemical potentials, density, temperature, field, and microscopic parameters.

6.5 Neutral species and anomalous magnetic moments

Neutral particles have $q = 0$ and therefore do not undergo Landau quantization in the minimal coupling approximation. Neutrons, however, possess magnetic moments and can couple to B through anomalous magnetic moment terms. Charged baryons can have both Landau and anomalous-moment effects. The two mechanisms should never be conflated.

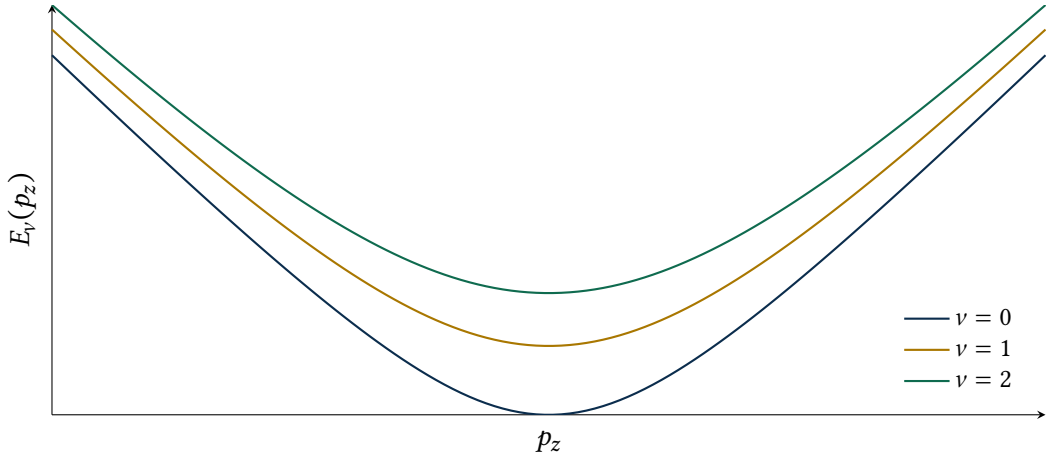


Figure 2: Schematic relativistic Landau spectrum in dimensionless units. Higher Landau levels have larger transverse energy gaps. The plot is illustrative, not a fit to any astrophysical source.

7 Thermodynamics and the Principal Stress Tensor

7.1 Total grand potential

Let

$$\Omega(\{\mu_i\}, T, B) = \Omega_m + \Omega_{\mathrm{HE}}^{\mathrm{ren}}, \tag{31}$$

where Ω_m contains the chosen dense-matter model and $\Omega_{\mathrm{HE}}^{\mathrm{ren}}$ is the renormalized vacuum contribution. The matter-sector scalar pressure is

$$P_m = -\Omega. \tag{32}$$

The magnetization is

$$M = -\left(\frac{\partial \Omega}{\partial B}\right)_{\{\mu_i\},T}. \tag{33}$$

7.2 Energy density

At zero temperature, the local energy density of the matter-plus-vacuum sector is

$$\varepsilon = \Omega + \sum_i \mu_i n_i + \frac{B^2}{2}. \tag{34}$$

The final term is the classical electromagnetic energy density. If SI units are desired, it becomes $B^2/(2\mu_0)$ and all corresponding conversions must be made consistently.

7.3 Principal pressures

With the local magnetic field aligned to the z-direction, the total principal stresses are

$$P_\parallel = P_m - \frac{B^2}{2}, \tag{35}$$

$$P_\perp = P_m - MB + \frac{B^2}{2}. \tag{36}$$

Therefore

$$\Delta P \equiv P_\perp - P_\parallel = B^2 - MB. \tag{37}$$

These equations are local rest-frame components of one stress-energy tensor; $P_\parallel$ and $P_\perp$ are not independent scalar equations of state.

7.4 Covariant form

Define the fluid four-velocity u^μ, projector $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$, and unit magnetic direction b^μ satisfying

$$u^\mu u_\mu = -1, \qquad b^\mu b_\mu = 1. \tag{38}$$

The stress tensor can then be written as

$$T^{\mu\nu} = \varepsilon u^\mu u^\nu + P_\perp h^{\mu\nu} + (P_\parallel - P_\perp) b^\mu b^\nu. \tag{39}$$

This is the geometrical form used by the global relativistic calculation.

7.5 Why transverse pressure need not vanish

The fact that $P_\perp$ can become small in a limiting regime does not mean that it must vanish for every large B. The expression contains competing terms involving matter pressure, magnetization, and classical field stress. The correct statement is conditional: a particular microscopic model may enter a regime where $P_\perp \ll P_\parallel$ or even where the total transverse stress becomes negative, but that behavior must be calculated and then checked for global equilibrium and stability.

No automatic filament theorem. A large B does not imply that all baryons become one-dimensional. The Landau level count is species-specific. Electrons can enter the lowest-Landau-level regime while neutrons remain non-Landau-quantized and continue to carry three-dimensional Fermi motion.

8 Species-by-Species Strong-Field Regimes

8.1 The lowest-Landau-level criterion

For a charged species i, the lowest Landau level (LLL) is the only occupied level when

$$\nu_{\max,i} = 0 \quad \Longleftrightarrow \quad \mu_i^{*2} - m_i^{*2} < 2|q_i|B. \tag{40}$$

This relation is exact within the minimal local spectrum. It immediately shows why a single field threshold cannot be attached to "matter" in general. The chemical potential, effective mass, charge, and interactions differ from species to species.

8.2 Electrons

Electrons are usually the first abundant charged species to exhibit strong Landau quantization because the electron mass is small. Their level spacing scales with $|eB|$, so for a given density they can have only a small number of occupied levels well before the baryonic species are similarly constrained. This can strongly modify charge neutrality, the electron chemical potential, and the composition of beta-equilibrated matter.

8.3 Protons and heavier charged species

Protons have the same charge magnitude but a much larger rest mass and a different chemical potential in interacting matter. Their Landau-level occupancy is therefore distinct from that of electrons. Hyper-

ons and other charged degrees of freedom inherit their own level structure. Any universal statement about "quantum filaments" must therefore be replaced by a composition-resolved analysis.

8.4 Neutrons

Neutrons are electrically neutral. Their energy spectrum does not acquire ordinary Landau levels, although their anomalous magnetic moment modifies the dispersion relation in a strong field. In a complete model, the neutron contribution can therefore remain strongly anisotropic without being one-dimensional in the Landau sense.

8.5 Composition feedback

The most interesting HMGQM feedback loop is not simply $B \to$ anisotropy. It is

$$B \to \text{Landau occupancy} \to \mu_i \to \text{composition} \to \Omega(\rho, B) \to M \to T_{\mu\nu} \to g_{\mu\nu} \to \text{equilibrium structure}. \tag{41}$$

Thus the field changes the matter composition, which changes the matter response to the field, which changes the stress-energy that curves spacetime.

8.6 Interactions and EOS closure

A serious calculation must provide θ_{micro} explicitly. For example, in a relativistic mean-field model the effective masses and vector potentials are solved self-consistently; in a tabulated EOS they may be provided as functions of density and composition; in a quark model the relevant thermodynamic potential follows from the chosen microscopic assumptions. HMGQM is therefore an architecture for coupling the microphysics to GR, not a replacement for the microscopic theory of QCD matter.

9 Axisymmetric General-Relativistic Equilibrium

9.1 Why spherical TOV is insufficient

A spherical TOV equation assumes spherical symmetry. A strong magnetic field explicitly selects a preferred direction and contributes a non-spherical stress-energy tensor. Therefore, once magnetic deformation matters, the correct equilibrium equations are partial differential equations in (r, θ) rather than one radial ordinary differential equation. A spherical anisotropic toy model can still be useful for intuition, but it is not the general HMGQM equilibrium system.

9.2 Metric ansatz

A standard stationary axisymmetric quasi-isotropic line element is

$$\mathrm{d}s^2 = -N^2\mathrm{d}t^2 + A^2\left(\mathrm{d}r^2 + r^2\mathrm{d}\theta^2\right) + C^2r^2\sin^2\theta(\mathrm{d}\phi - \omega\mathrm{d}t)^2, \tag{42}$$

where N, A, C, ω are functions of (r, θ). For a non-rotating configuration $\omega = 0$. For a rotating star the frame-dragging term is retained.

9.3 Magnetic potential

An axisymmetric poloidal magnetic field can be represented using an azimuthal potential component $A_\phi(r, \theta)$. Rotation and electric fields may require a nonzero A_t. The exact current distribution is part of the model. One does not obtain a unique global magnetic field by specifying only a central scalar value of B.

9.4 Coupled boundary-value problem

The complete stationary system is

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{43}$$
$$\nabla_\mu H^{\mu\nu} = J^\nu, \tag{44}$$
$$\nabla_{[\alpha} F_{\beta\gamma]} = 0, \tag{45}$$
$$\nabla_\mu T^{\mu\nu} = 0, \tag{46}$$

plus the microscopic EOS, charge neutrality, beta equilibrium, thermal state, current prescription, and boundary conditions.

9.5 Relativistic force balance

Projecting stress-energy conservation orthogonally to u^μ gives the relativistic spatial force-balance equation

$$h^\alpha_{\ \nu} \nabla_\mu T^{\mu\nu} = 0. \tag{47}$$

Because $T_{\mu\nu}$ already contains the electromagnetic and magnetization contributions derived from the action, no extra Lorentz-force term may be inserted twice by hand.

9.6 Boundary conditions

The stellar interior must be matched to a regular exterior solution. At the surface the pressure normal to the surface must satisfy the appropriate stress condition, the electromagnetic field must satisfy Maxwell boundary conditions, and the metric must join smoothly to the exterior spacetime. For a realistic magnetar, the exterior may be treated as a magnetosphere rather than an exact vacuum; whichever approximation is chosen must be stated.

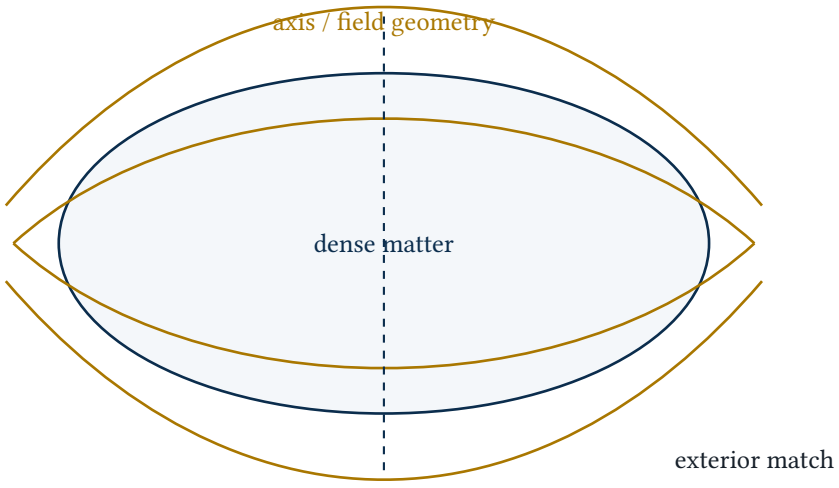


Figure 3: Schematic axisymmetric HMGQM configuration. The interior matter distribution and magnetic geometry are solved together; the figure is not a claim of a particular field topology.

10 Magnetic Geometry, Energetics, and Deformation

10.1 Geometry is a dynamical variable

The local scalar $B = |\mathbf{B}|$ is not enough to determine the global equilibrium. Two stars with identical field amplitudes can have different Lorentz forces and different quadrupole moments if one is predominantly poloidal and the other has a strong toroidal component. The complete framework therefore treats field topology as part of the physical model.

10.2 Magnetic energy scale

A useful order-of-magnitude diagnostic is the magnetic energy

$$E_B \sim \int \mathrm{d}^3x \, \frac{B^2}{2}. \tag{48}$$

For a uniform-field toy estimate of a star of radius R,

$$E_B \sim \frac{B^2}{2} \frac{4\pi R^3}{3}. \tag{49}$$

Taking $B \sim 10^{14}\,\mathrm{T}$ and $R \sim 12\,\mathrm{km}$ gives an energy scale of order 10^{46} J. This is comparable to the characteristic gravitational binding-energy scale of a canonical neutron star, illustrating why extremely strong internal fields cannot be treated as a tiny correction. The estimate is intentionally a scaling argument; stable stars do not possess perfectly uniform fields, and actual equilibrium energies are geometry-dependent.

10.3 Ellipticity

Magnetic stresses can generate a mass quadrupole. Define the dimensionless ellipticity schematically as

$$\epsilon_B = \frac{I_1 - I_2}{I_3}, \tag{50}$$

with principal moments of inertia defined relative to the rotation axis. The sign and magnitude depend strongly on field topology. Thus a measured or constrained ellipticity can test a model only after the interior field configuration has been specified.

10.4 Mass-radius response

A sequence of solutions yields

$$M(R; \theta_{\mathrm{micro}}, \mathscr{T}_B, \Omega), \tag{51}$$

where $\mathscr{T}_B$ denotes field topology and Ω the rotation rate. The magnetized sequence should be compared with a baseline sequence computed using the same microscopic EOS and the same external assumptions. A mass shift is then a model effect, not automatically evidence for a new phase.

10.5 Magnetically supported configurations

Prior fully relativistic calculations show that very strong magnetic fields can deform compact stars and alter maximum supported masses. Recent work also finds magnetically supramassive and hypermassive equilibrium sequences for sufficiently strong mixed fields. HMGQM therefore does not claim the mere existence of magnetic support as its novelty. The question is whether the additional strong-field QED and microscopic quantization effects produce a distinctive regime beyond the known magnetized-star solutions.

11 Stability, Causality, and Energy Conditions

11.1 Hierarchy of stability checks

The original "stable if and only if" inequalities are replaced by a hierarchy. A physically acceptable solution should first satisfy local thermodynamic consistency, then global equilibrium checks, and finally a dynamical perturbation test.

11.2 Local thermodynamic consistency

Along a specified thermodynamic path, the sound speed satisfies the causality requirement

$$0 \leq c_s^2 = \left(\frac{\partial P}{\partial \varepsilon}\right)_{\text{path}} \leq 1. \tag{52}$$

The magnetized system is anisotropic, so the physically relevant characteristic speeds must be computed from the full anisotropic magnetofluid equations rather than by blindly assigning one scalar c_s. Positive compressibility and appropriate susceptibilities are necessary local checks.

11.3 Energy conditions

The weak, dominant, and null energy conditions can be evaluated from the total $T_{\mu\nu}$. For an anisotropic diagonal stress in the comoving frame, one examines conditions involving ε, $P_{\parallel}$, and $P_{\perp}$. Violation is not mathematically forbidden in every effective quantum system, but any violation must be identified and physically interpreted rather than hidden.

11.4 Global sequence tests

For a family of equilibria, track gravitational mass M, baryon mass M_b, angular momentum J, central density ρ_c, magnetic energy E_B, and other control parameters. Turning points can identify candidate changes in stability, but a turning-point argument alone is not a proof of dynamical stability for an axisymmetric magnetized system.

11.5 Dynamical perturbations

The decisive test is the linearized coupled system

$$\delta G_{\mu\nu} = 8\pi G\, \delta T_{\mu\nu}, \tag{53}$$

$$\delta(\nabla_\mu H^{\mu\nu} - J^\nu) = 0, \tag{54}$$

$$\delta(\nabla_\mu T^{\mu\nu}) = 0. \tag{55}$$

For perturbations behaving as $e^{-i\omega t}$, instability corresponds to $\operatorname{Im}\omega > 0$. The mode problem may contain fluid, electromagnetic, gravitational, and mixed modes.

Stability claim rule. HMGQM v3.0 does not label a star "stable" merely because $P_\perp \geq 0$ or $\partial P/\partial \rho > 0$. Those are local checks. A full stability statement requires the eigenvalue problem or another mathematically justified dynamical criterion.

12 What H-Matter Means in the Complete Theory

12.1 Operational regime definition

A region is labelled H-Matter when

$$\chi_B \gg 1, \qquad n_B \gtrsim n_{\rm sat}, \qquad \nu_{{\rm max},i} = O(1) \tag{56}$$

for at least one dynamically important charged species and when electromagnetic/QED contributions make an order-unity contribution to the relevant local stress or global equilibrium. This definition is deliberately operational.

12.2 Regime versus phase

A change from many Landau levels to the lowest Landau level can be smooth. Anisotropy can likewise grow continuously with B. Therefore neither effect is sufficient to establish a thermodynamic phase transition. A new phase requires, for example, a nonanalyticity in the thermodynamic potential or an order parameter Φ with a well-defined transition structure.

A formal criterion can be written as

$$\frac{\partial^2 \Omega}{\partial B^2} \text{ or another susceptibility} \tag{57}$$

showing a nonanalytic feature, or

$$\Phi = 0 \text{ outside the phase}, \qquad \Phi \neq 0 \text{ inside the phase}, \tag{58}$$

with the appropriate non-continuous or divergent behavior at the phase boundary. Until such a result is derived, "H-Matter" is a scientifically safe name for a high-field/high-density regime, not proof of a new fundamental substance.

12.3 Why this is still a meaningful hypothesis

The operational regime can be physically important even if no new phase transition exists. It can alter composition, magnetization, anisotropic stress, global deformation, photon propagation, and oscillation spectra. A theory does not need a phase transition to make a new calculation scientifically useful.

12.4 Relation to known magnetized matter

Existing work already treats anisotropic pressure in dense magnetized matter, including cases in which magnetic fields drive strong directional stress and possible instabilities. The HMGQM hypothesis is therefore not "anisotropy exists." The stronger question is whether the combined high-field QED + dense-matter + GR system has a robust domain in which the coupled response differs enough from baseline magnetized EOS models to deserve a distinct effective classification.

13 Observable Consequences: Photon Propagation

13.1 Vacuum birefringence

Strong-field QED predicts polarization-dependent photon propagation in a magnetized vacuum. This is an established QED effect and is not presented here as a unique discovery of HMGQM. The HMGQM-specific calculation is more demanding: compute the photon propagation through the self-consistent magnetic field and curved spacetime of the equilibrium solution and compare it with a baseline magnetized-star model.

For finite-field calculations, the refractive response should be evaluated using the same finite-field

QED sector that controls the stress-energy, rather than extending a weak-field expression far beyond its regime. This is particularly important near the stellar surface of very strongly magnetized objects.

13.2 Polarization transport

For an observer at infinity, the measurable quantities include the energy-dependent polarization degree, polarization angle, phase dependence across a rotation cycle, and changes in these observables caused by vacuum resonances or mode conversion. The geometry of the magnetic axis relative to the observer matters. So does the atmospheric composition and magnetospheric plasma state.

HMGQM photon test = self-consistent field geometry+curved-spacetime ray tracing+finite-field QED transport−ma (59)

13.3 No "polarized gravitational lensing" claim

The correct terminology is *polarization-dependent photon propagation in curved magnetized spacetime.* The polarization dependence is fundamentally an electrodynamic propagation effect. Gravity curves the rays and redshifts the photons, while QED supplies the birefringent response.

13.4 2026 observational context

Recent phase- and energy-resolved X-ray polarization measurements of the radio magnetar 1E 1547.0-5408 reported high polarization degrees and patterns consistent with vacuum-birefringence-governed magnetospheric propagation. This is an important observational milestone for strong-field QED. It does not by itself establish H-Matter. Instead, it provides a realistic data channel through which the HMGQM strong-field transport calculation can be confronted with a baseline model.

14 Observable Consequences: Gravitational Waves

14.1 Mode spectrum from perturbation theory

The complete theory does not assume a frequency correction of the form $\omega = \omega_0 + \delta\omega(B, \Delta P)$ as a final result. Instead, one linearizes the full axisymmetric equilibrium and solves

$$\mathscr{D}[\delta X; g_{\mu\nu}, F_{\mu\nu}, \Omega] = 0, \tag{60}$$

where δX includes matter, electromagnetic, and metric perturbations. The eigenvalues $\omega_{n\ell m}$ then become model predictions.

14.2 Magnetic splitting

A magnetic field can break degeneracies that exist in spherical stars. Axisymmetric deformation changes the coupling among angular harmonics, while magnetic stresses introduce Alfvén-like and mixed magneto-gravitational modes. Potential observables include frequency shifts, mode splittings, damping times, and polarization content of the gravitational radiation.

14.3 Post-merger caution

GW170817 demonstrated that gravitational waves can constrain the neutron-star equation of state through tidal effects, while searches also place limits on a possible post-merger signal. However, a detection of a merger does not automatically provide a measured post-merger oscillation frequency. The original HMGQM plot that treated GW170817 and GW190814 as direct frequency points is there-

fore removed from the complete theory.

14.4 What the theory should predict

A useful prediction is a differential quantity:

$$\Delta\omega_{n\ell m} = \omega_{n\ell m}^{\mathrm{HMGQM}} - \omega_{n\ell m}^{\mathrm{baseline}}, \tag{61}$$

where the baseline uses the same mass, rotation, surface field, and microscopic EOS but omits the specific HMGQM strong-field QED/Landau coupling under test. Such a matched comparison is much harder to dismiss as parameter retuning.

15 Observable Consequences: Mass, Radius, Tidal Response

15.1 Mass-radius sequences

A numerical equilibrium survey produces families such as

$$M(R), \quad R(M), \quad M_{\max}, \quad R_{1.4}, \quad R_{2.0}, \tag{62}$$

for specified EOS and field topology. The correct comparison is always like-for-like: the same microphysical EOS, same thermal prescription, same rotation state, and matched exterior assumptions.

15.2 Tidal deformability

The dimensionless tidal deformability Λ is related to the Love number and compactness through

$$\Lambda = \frac{2}{3} k_2 C^{-5}, \qquad C = \frac{GM}{Rc^2} \tag{63}$$

in the spherical baseline. Strong magnetic deformation can require a generalized axisymmetric treatment, but the spherical expression remains a useful reference point. Gravitational-wave measurements of tidal effects therefore offer a natural comparison channel.

15.3 Moment of inertia

The moment of inertia $I(M)$ is sensitive to compactness and internal structure. In a rotating axisymmetric configuration the exterior spacetime provides a relativistic relation among M, J, and the effective moment of inertia. A magnetic correction is therefore observable in principle through precise timing or multimessenger inference, though actual detectability depends on field geometry and systematics.

15.4 Ellipticity and continuous waves

A non-axisymmetric component of the mass quadrupole can generate continuous gravitational radiation from a rotating star. The HMGQM framework is not claiming a specific amplitude before the field topology is solved. Instead, it predicts a field-configuration-dependent ellipticity that can be propagated to an expected strain and compared with detector upper limits.

16 Electromagnetic Observables and Magnetospheric Coupling

16.1 Radiative transfer

The interior theory alone does not predict an X-ray spectrum. One needs a surface atmosphere model, a magnetospheric plasma prescription, and relativistic ray tracing. HMGQM can supply the spacetime

and field geometry, while the radiative transfer module determines how those conditions map to observables.

16.2 Potential discriminants

The most useful electromagnetic diagnostics are not raw luminosities, which are model-dependent, but differential quantities sensitive to geometry and QED response: phase-resolved polarization angle, polarization degree, vacuum-resonance features, energy-dependent pulse profiles, and correlations between polarization and inferred field orientation.

16.3 Magnetosphere versus interior

A major source of degeneracy is that an observer sees radiation after it has propagated through both the surface atmosphere and the magnetosphere. Therefore an interior-only model cannot be compared directly to data without specifying the propagation environment. HMGQM v3.0 treats the interior as one layer of a multimodule inference chain.

16.4 Specific HMGQM task

The theory is most informative when two models are matched at the surface:

1. baseline magnetized compact-star model;
2. HMGQM strong-field effective model.

The quantity to compare is then the difference in predicted polarization/radiative observables after the same nuisance parameters are marginalized.

17 Observational Discipline: What Current Data Do and Do Not Say

17.1 GW170817

GW170817 was the first binary-neutron-star gravitational-wave event with an electromagnetic counterpart. Its inspiral waveform constrained tidal deformability and, together with other information, narrowed the viable neutron-star equation-of-state space. It is therefore an important baseline for HMGQM, but it is not direct evidence that H-Matter exists.

17.2 GW190814

GW190814 contained a compact object of about 2.6 $M_\odot$ orbiting a roughly 23 $M_\odot$ black hole. The nature of the secondary remains uncertain in the standard interpretation: it could be a heavy neutron star or a light black hole. HMGQM may investigate whether strongly magnetized matter can alter the maximum mass of compact stars, but the event should not be treated as a direct measurement of an H-Matter oscillation mode.

17.3 NICER

NICER pulse-profile modelling has produced mass-radius information for massive pulsars such as PSR J0740+6620. Such measurements provide a direct observational handle on the global structure of compact stars. In the HMGQM program, they constrain the allowed mass-radius sequence after magnetic geometry and observational nuisance parameters are included.

17.4 IXPE and the 2026 magnetar result

X-ray polarization is a particularly relevant channel because strong-field QED directly affects photon propagation. The 2026 Nature result for 1E 1547.0-5408 provides a contemporary motivation for treating vacuum birefringence as an observationally accessible effect rather than a purely formal correction.

Observation rule. No observation is labelled "evidence for H-Matter" unless the same quantity cannot be reproduced by a baseline magnetized-star calculation after fitting the ordinary nuisance parameters. A successful HMGQM fit alone is insufficient; the discriminating power comes from model comparison.

18 Dimensionless Regime Map

18.1 Why dimensionless diagnostics matter

A theory that combines gravity, QED, and dense matter spans enormous ranges of scales. Dimensionless variables expose the relevant ordering without committing the theory to one unit system. Four quantities are especially useful: χ_B, β_B, $\nu_{\max}$, and χ_R.

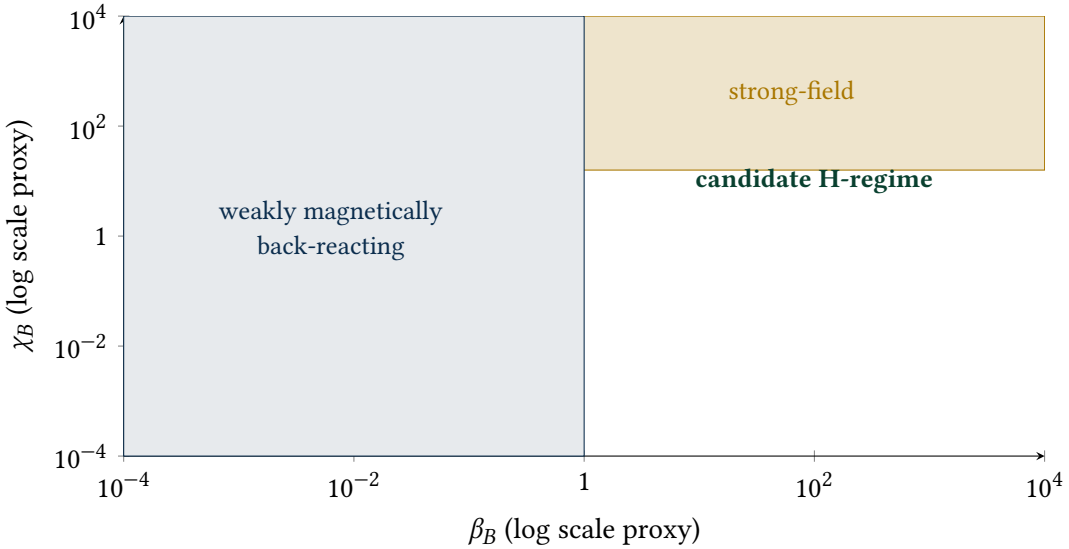


Figure 4: Schematic, dimensionless regime map. It illustrates the logic of the diagnostics rather than a fitted phase diagram. Additional conditions such as Landau-level occupancy and one-loop control must also be satisfied.

18.2 Regime taxonomy

For numerical studies it is useful to classify a local state as one of four categories:

1. weak-field ordinary dense matter: $\chi_B \lesssim 1$ and $\beta_B \ll 1$;
2. strongly quantized matter: $\chi_B \gg 1$ and at least one important charged species has small $\nu_{\max}$, but magnetic stress remains modest;
3. magnetically back-reacting matter: β_B is not negligible and global geometry is field-controlled;
4. H-regime candidate: strong quantization and strong magnetic back-reaction coexist while the effective-theory control criteria remain valid.

This taxonomy prevents the theory from assigning the same physical label to qualitatively different situations.

19 Limiting Cases and Recovery of Established Physics

19.1 Zero-field limit

As $B \to 0$, Landau sums must approach the corresponding three-dimensional momentum integrals and the magnetization must vanish:

$$M \to 0, \qquad P_{\parallel} - P_{\perp} \to 0. \tag{64}$$

The field-corrected matter EOS must approach the chosen unmagnetized EOS.

19.2 Weak-field QED limit

As $B/B_{\mathrm{QED}} \to 0$, the proper-time Heisenberg-Euler sector must reduce to its known weak-field expansion. This is a stringent test of signs, renormalization conventions, and numerical implementation.

19.3 Many-Landau-level limit

When $\nu_{\max} \gg 1$, the discrete Landau sum must converge toward the continuum limit. A numerical code should demonstrate this convergence explicitly as a resolution test.

19.4 Weak magnetic stress

When $B^2/2 \ll P_m$ and the QED correction is negligible, the global equilibrium should continuously approach the ordinary relativistic star. In this limit, spherical symmetry is recovered to leading order if the boundary conditions are also spherical.

19.5 Newtonian limit

For $C = GM/(Rc^2) \ll 1$ and small velocities, the Einstein equations should reduce to Newtonian gravity plus the appropriate magnetostatic force balance. This provides an independent code check because the resulting limit can be implemented with a simpler solver.

Limiting-case theorem of the framework. HMGQM v3.0 is acceptable only if every numerical realization reproduces known limits as the corresponding control parameters are relaxed. A model that works only in the extreme regime but fails its weak-field, many-level, or Newtonian limits is not an acceptable HMGQM realization.

20 Numerical Implementation Blueprint

20.1 Stage 1: microphysical table

The first module constructs a thermodynamically consistent table or callable function for

$$\Omega_m(\{\mu_i\}, T, B; \theta_{\mathrm{micro}}), \tag{65}$$

and its derivatives. The implementation should expose n_i, M, ε, $P_{\parallel}$, and $P_{\perp}$ and verify Maxwell relations numerically.

20.2 Stage 2: equilibrium solver

The second module solves the axisymmetric Einstein-Maxwell boundary-value problem. A pseudo-spectral or high-order finite-difference method is appropriate. The solver should treat the metric potentials, electromagnetic potentials, and matter variables simultaneously.

20.3 Stage 3: global diagnostics

For each converged model, store

$$\{M, M_b, J, E_B, R_{\rm eq}, R_{\rm pole}, \epsilon_B, \rho_c, B_c, \ldots\}. \tag{66}$$

This permits systematic scans over central density, field strength, rotation rate, and topology.

20.4 Stage 4: perturbations

The perturbation solver linearizes the equilibrium equations and computes eigenfrequencies, damping times, and mode couplings. The perturbation system should reuse the same EOS derivatives as the equilibrium code.

20.5 Stage 5: observational layer

The final layer performs relativistic ray tracing, polarized radiative transfer, waveform generation, or other observable modelling. The model output should be compared to baseline models with the same nuisance parameters.

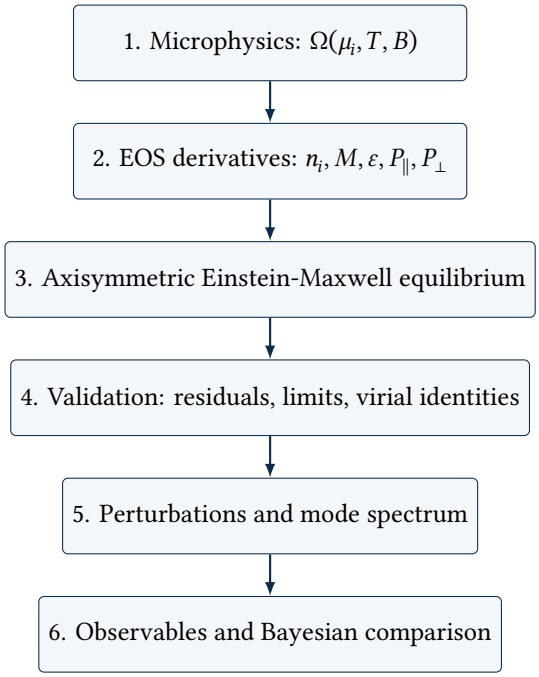


Figure 5: Numerical pipeline. No observational claim should be made before the equilibrium solution passes the mathematical validation stage.

21 Validation Suite

21.1 Thermodynamic identities

Finite-difference derivatives of Ω should reproduce

$$n_i = -\frac{\partial \Omega}{\partial \mu_i}, \qquad M = -\frac{\partial \Omega}{\partial B}, \tag{67}$$

and the pressure identities used by the stress tensor. This catches a large class of implementation errors before GR is introduced.

21.2 Landau-sum convergence

For fixed density and chemical potential, increase the number of Landau levels until observables converge. In the many-level limit, compare against the unquantized continuum result.

21.3 Einstein-equation residuals

A converged model should have small residuals in all independent metric equations and Maxwell equations. The residual norm should be reported rather than merely stating that the solver converged.

21.4 Conservation residual

Numerically evaluate

$$\mathscr{R}_T^{\nu} = \nabla_\mu T^{\mu\nu} \tag{68}$$

and show that it converges toward zero with numerical resolution. This is one of the strongest internal checks because it tests the consistency of the combined matter-plus-field stress tensor.

21.5 Virial and integral identities

Where available, relativistic virial identities and Komar/ADM mass consistency checks should be used. Magnetized rotating-star codes already use such global consistency tests; an HMGQM implementation should do the same.

21.6 Mesh and gauge independence

A final publication-quality computation should demonstrate convergence under changes in resolution, coordinate mapping, and electromagnetic gauge representation. A physical observable must not depend on the gauge used to represent A_μ.

22 Falsifiability and Failure Conditions

22.1 Theory-level failure

The framework fails at the theory level if it cannot produce a well-posed thermodynamic potential, violates its own limiting cases, or produces a total stress tensor inconsistent with the action.

22.2 Model-level failure

A specific microphysical realization fails if it predicts negative susceptibilities in a region assumed stable, violates causality, fails charge neutrality or beta equilibrium, or produces no self-consistent global solution under its claimed conditions.

22.3 Observational failure

HMGQM becomes empirically disfavored if a baseline magnetized-star model reproduces all relevant observations while the HMGQM-specific corrections are inconsistent with the same data. The frame-

work is most strongly testable when it predicts correlated signatures across independent channels.

22.4 Phase-claim failure

If extensive calculations show that the H-regime always evolves smoothly from ordinary magnetized matter with no phase transition, the scientific conclusion should be that H-Matter is a useful regime classification rather than a new thermodynamic phase. This is not a defeat of the framework; it is a falsification of one possible interpretation.

The strongest falsifiable statement. There exists a domain of compact-object parameter space in which the self-consistent strong-field QED + Landau-quantized dense-matter + Einstein-Maxwell model produces observables that cannot be reproduced by a matched baseline model within its uncertainty budget. If this statement fails, the extra HMGQM classification has no independent empirical content.

23 Uncertainty, Degeneracies, and Model Comparison

23.1 Microscopic uncertainty

Dense-matter microphysics is uncertain at supranuclear density. A result that depends strongly on one arbitrarily chosen EOS is not a robust prediction. The correct procedure is to propagate EOS uncertainty through the HMGQM calculation, either by an ensemble of microscopic models or by a parameterized posterior.

23.2 Field topology uncertainty

Poloidal, toroidal, and mixed fields can produce different deformations at comparable local B. Therefore a claimed observational signature must either be topology-robust or explicitly report how it changes with topology.

23.3 Rotation and inclination

Observed polarization and pulse profiles depend on the inclination of the spin axis and magnetic axis relative to the observer. Similarly, gravitational-wave mode amplitudes depend on viewing geometry. These parameters must be marginalized rather than fixed to produce a desired fit.

23.4 Bayesian comparison

Let D denote the data and θ the nuisance parameters. The relevant comparison is a model evidence or likelihood ratio between a baseline model $\mathscr{M}_0$ and HMGQM $\mathscr{M}_H$:

$$\mathscr{B}_{H0} = \frac{p(D|\mathscr{M}_H)}{p(D|\mathscr{M}_0)}. \tag{69}$$

A high likelihood at the cost of much greater parameter freedom does not automatically make HMGQM preferable. The extra degrees of freedom must buy predictive power.

23.5 Sensitivity budget

Every claimed effect should be accompanied by a sensitivity budget showing which uncertainty dominates: EOS, field topology, atmosphere, distance, instrument calibration, or QED truncation. The purpose is to prevent a small formal correction from being presented as an observable discovery when astrophysical systematics are much larger.

24 Comparison with Baseline Physics

Aspect	Baseline magnetized compact star	HMGQM complete framework
Gravity	General relativity	General relativity
Matter	Chosen dense-matter EOS	Same class of EOS, evaluated with strong-field response
Charged species	Landau quantization may be included	Landau quantization is mandatory in the H-regime
QED vacuum	Optional/finite-field propagation treatment	One-loop Heisenberg-Euler sector included when valid
Stress tensor	Matter + EM + chosen corrections	Single action-derived total $T_{\mu\nu}$
Geometry	Axisymmetric if field back-reaction matters	Axisymmetric by construction in the general case
Phase claim	Not assumed	Requires nonanalyticity or order parameter
Observables	M, R, Λ, polarization, modes	Same, plus explicit matched HMGQM-minus-baseline differences

Table 1: Conceptual comparison. The HMGQM novelty is the coupled high-field regime and its predictions, not ownership of the individual physical ingredients.

24.1 Why the comparison is scientifically necessary

A new name can make a familiar calculation sound novel unless the baseline is explicit. This table therefore serves as a methodological guardrail. If a result can be reproduced by existing magnetized-star calculations with the same microphysical inputs, the result belongs to the baseline rather than to HMGQM specifically.

24.2 What would count as genuine novelty

A strong novelty claim would require at least one of the following: a new stable branch caused by the coupled strong-field QED/matter feedback; a robust nonanalytic phase transition not present in the baseline model; a qualitatively new mode family or splitting pattern; or a correlated observational signature that cannot be reproduced by retuning ordinary magnetic-field geometry and EOS parameters.

25 A Worked Scaling Example: When Magnetic Stress Matters

Consider a canonical compact object with $M = 1.4M_\odot$ and $R = 12$ km. The Newtonian gravitational binding-energy scale is

$$E_G \sim \frac{GM^2}{R}. \tag{70}$$

The magnetic energy of a uniform-field toy model is

$$E_B \sim \frac{B^2}{2\mu_0}\frac{4\pi R^3}{3} \tag{71}$$

in SI units. Evaluating the ratio gives

$$\frac{E_B}{E_G} \sim \frac{2\pi B^2 R^4}{3\mu_0 G M^2}. \tag{72}$$

This ratio scales as B^2. The worked example therefore demonstrates a central lesson: there exists a field strength at which magnetic stress is no longer a small correction to the star's gravitational structure.

This calculation does not prove that such a uniform field is physically realized. It does not specify current support, stability, or topology. Its role is diagnostic: it explains why a fully relativistic treatment becomes necessary at sufficiently strong fields, in agreement with the known magnetized-star literature.

25.1 From scaling to HMGQM

The HMGQM step occurs when the magnetic back-reaction condition is joined by strong Landau quantization and a finite-field QED response. The relevant regime is therefore an intersection of conditions, not a single threshold:

$$\boxed{\text{H-regime} = \text{high density} \cap \text{strong quantization} \cap \text{magnetic back-reaction} \cap \text{QED control.}} \tag{73}$$

26 The Complete Master System

26.1 Compact statement

The full HMGQM v3.0 framework is summarized by the following system:

$$\begin{aligned}
S &= \int \mathrm{d}^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{4}F^2 + \Delta\mathscr{L}_{\mathrm{HE}} + \mathscr{L}_{\mathrm{matter}}\right],\\
G_{\mu\nu} &= 8\pi G T_{\mu\nu},\\
\nabla_\mu H^{\mu\nu} &= J^\nu,\\
\nabla_{[\alpha}F_{\beta\gamma]} &= 0,\\
\nabla_\mu T^{\mu\nu} &= 0,\\
\Omega &= \Omega_m + \Omega_{\mathrm{HE}}^{\mathrm{ren}},\\
M &= -\partial\Omega/\partial B,\\
\varepsilon &= \Omega + \sum_i \mu_i n_i + B^2/2,\\
P_\parallel &= -\Omega - B^2/2,\\
P_\perp &= -\Omega - MB + B^2/2,\\
n_i &= \frac{|q_i|B}{2\pi^2}\sum_\nu g_\nu k_{F,i\nu},\\
E_{i\nu}^2 &= p_z^2 + m_i^{*2} + 2\nu|q_i|B,\\
\nu_{\max,i} &= \left\lfloor \frac{\mu_i^{*2} - m_i^{*2}}{2|q_i|B} \right\rfloor.
\end{aligned}$$

The global solution is obtained with the stationary axisymmetric metric of equation (42), a specified magnetic topology, a stated microscopic closure, and explicit boundary conditions. The observable prediction is not any one line of this system; it is the output of a validated equilibrium-plus-perturbation solution.

26.2 Minimal parameter set

A practical model can be specified by

$$\Theta = \{\theta_{\text{micro}}, B_c, \mathcal{T}_B, \Omega, T, \text{composition rule, surface model}\}. \tag{74}$$

For each Θ, solve the equilibrium and compute derived observables. The parameter vector should never include both a thermodynamic pressure and an independent magnetic-pressure correction, because that would duplicate physics already encoded in the action.

26.3 Definition of a successful model realization

A successful HMGQM realization is one that:

1. is internally thermodynamically consistent;
2. converges numerically;
3. satisfies Einstein-Maxwell and conservation residuals;
4. recovers known limiting cases;
5. obeys its stated validity domain;
6. produces at least one clearly defined, falsifiable observable difference from a matched baseline.

27 Predictions, Non-Predictions, and Research Claims

27.1 Predictions the framework is designed to make

The following are legitimate target predictions after numerical solution:

- magnetic-field-dependent mass-radius sequences;
- field-topology-dependent deformation and ellipticity;
- composition changes driven by Landau occupancy;
- finite-field QED corrections to polarization transport;
- changes in mode frequencies and magnetic splitting;
- correlated changes among M, R, I, Λ, and polarization observables.

27.2 Claims it does not make

The complete formulation intentionally does not claim:

- a complete theory of quantum gravity;
- a universal new particle or new fundamental field;
- laboratory synthesis of H-Matter;
- that every $B \gg B_{\text{QED}}$ object contains a new phase;
- that GW170817 or GW190814 directly measured an HMGQM mode;
- that anisotropy alone proves a phase transition.

27.3 Why negative statements matter

A personal theoretical framework becomes more useful when it states where it is expected to fail. The boundaries above are not rhetorical disclaimers; they are part of the model definition. A numerical code that crosses them must report that the calculation is outside the theory's intended validity region.

28 Research Program: From Theory to a Publishable Test

28.1 Phase I – microphysical closure

Select one concrete EOS and compute $\Omega_m(\mu_i, B)$, including Landau quantization and anomalous magnetic moments where appropriate. Validate against known unmagnetized and weak-field results.

28.2 Phase II – axisymmetric equilibrium

Implement the full stationary Einstein-Maxwell solver with a self-consistent field topology. Demonstrate convergence, conservation, and virial consistency. Produce a sequence of models spanning from weak to strong magnetic back-reaction.

28.3 Phase III – H-regime survey

Compute the diagnostics χ_B, β_B, $\nu_{\max}$, κ_{loop}, and χ_R throughout each model. Map where the H-regime conditions overlap. This produces a physical parameter-space map rather than a purely verbal regime definition.

28.4 Phase IV – perturbation spectrum

Linearize the equilibrium equations and compute stable and unstable modes. Determine whether magnetic/QED effects produce a mode family or splitting pattern that survives changes of EOS and topology.

28.5 Phase V – multimessenger comparison

Generate mass-radius, tidal, polarization, and gravitational-wave observables. Compare each against a matched baseline using the same priors and nuisance parameters.

28.6 Phase VI – phase test

Search the microscopic thermodynamic potential for nonanalytic structure. If a true phase transition exists, calculate its coexistence conditions, latent heat or susceptibility behavior, and macroscopic signatures. If not, retain "H-Matter" as a regime label.

29 Loophole Closure: Complete Repair Map

Original vulnerability	Why it was a problem	v3.0 resolution
Planck-scale language with classical GR	Internal domain contradiction	Classical GR is explicit; quantum gravity is outside scope; curvature diagnostic added.
Weak-field F^4 QED term used at large B	Extrapolated a low-field expansion	Finite-field one-loop Heisenberg-Euler proper-time form used.

3D Fermi integral labelled Landau-quantized	Missing magnetic density of states	Exact Landau-level sum inserted for charged species.
All baryons assumed one-dimensional	Neutral species do not Landau-quantize	Species-by-species occupancy plus AMM treatment.
Magnetic/vacuum stress added twice	Double counting in $T_{\mu\nu}$	One action and one thermodynamic potential define all stresses.
Spherical anisotropic TOV after axial symmetry claim	Spherical TOV cannot represent general field geometry	Stationary axisymmetric Einstein-Maxwell PDE system.
"Stable iff" local inequalities	Local conditions are not global dynamical proof	Stability hierarchy ending in eigenmode analysis.
"Polarized gravitational lensing"	Mislabels the physical mechanism	Polarization-dependent photon propagation in curved magnetized spacetime.
GW events used as frequency points	Observations do not provide those plotted values	Removed; events used only for their actually measured constraints.
New phase inferred from anisotropy	Anisotropy can be a crossover	Phase requires nonanalyticity or order parameter.
Undefined effective mass $m^*(\rho, B)$	Hidden microscopic assumption	m^* belongs to explicit microphysical closure.
Rotation shown but not in equations	Diagram outran mathematics	Rotation enters the stationary metric through ω.
Magnetic field treated as scalar input	Global force depends on geometry/topology	A_ϕ / current prescription treated as dynamical model data.

30 Reference-Linked Research Map

This section makes the document's external links explicit. The link labels are intentionally attached to the scientific keywords rather than printed as raw web addresses.

- **Heisenberg-Euler effective Lagrangians**: pedagogical review of the one-loop QED effective action and extensions.
- **Heisenberg-Euler effective action, 75 years on**: historical and technical review of the full nonperturbative one-loop action.
- **Einstein-Maxwell rotating magnetized stars**: early fully relativistic self-consistent axisymmetric solutions.
- **anisotropic pressure in strong magnetic fields**: dense neutron-matter pressure anisotropy under strong fields.
- **magnetized QCD pressure anisotropy**: field-dependent parallel and transverse pressure in dense QCD matter.
- **chiral effective field theory and high-density EOS**: controlled nuclear-theory input and uncertainty discussion.

- **magnetically supramassive and hypermassive compact stars**: contemporary fully relativistic magnetic support calculations.
- **2026 magnetar vacuum-birefringence observation**: recent X-ray polarization result relevant to strong-field QED transport.
- **finite-field QED corrections**: contemporary finite-field treatment of vacuum birefringence and magnetar polarization transport.
- **GW170817**: official event record and documentation.
- **GW190814**: official event record and documentation.
- **NICER PSR J0740+6620 radius update**: updated high-mass neutron-star radius analysis.

31 Note About the Author

Anish Kushwaha

Independent student / researcher

Anish Kushwaha presents HMGQM as a curiosity-driven theoretical research project focused on the intersection of extreme astrophysics, theoretical physics, computation, and system-level reasoning. This document is intentionally framed as an exploratory research manuscript rather than a claim of completed experimental discovery. The author's goal is to turn an initial conceptual hypothesis into a mathematically explicit framework that can survive critical examination, numerical testing, and observational comparison.

Basic contact and public links

Portfolio • Email • X.com

Attribution note. The linked contact details above are intentionally presented as hyperlinks only, as requested. The document does not print the raw web addresses or mail address.

32 Conclusion

HMGQM v3.0 is the complete, narrowed formulation of the original idea. Its central proposal is no longer that a mysterious quantum-gravitational substance automatically appears at Planck-scale curvature. Instead, it identifies a physically meaningful regime in which strong magnetic fields, quantum electrodynamic vacuum response, charged-particle Landau quantization, dense-matter microphysics, and relativistic gravity must be treated together.

The mathematical core is now unambiguous. One effective action defines the coupled system. The Heisenberg-Euler sector is retained in finite-field form rather than replaced by an inappropriate weak-field truncation. Charged fermions are quantized with the correct Landau-level density of states. Neutral particles and anomalous magnetic moments are treated separately. Magnetization and principal stresses are derived from the same thermodynamic potential that feeds the stress-energy tensor. The global equilibrium is axisymmetric and governed by the coupled Einstein-Maxwell equations, with magnetic topology and boundary conditions made explicit.

The word "phase" has also been put on a strict scientific footing. H-Matter is a regime label until a thermodynamic nonanalyticity or an independently derived order parameter demonstrates a genuine phase transition. This distinction prevents the theory from claiming more than the mathematics can establish while preserving the original physical intuition.

The observational program is deliberately comparative. Vacuum birefringence is treated as established strong-field QED physics, while HMGQM asks whether a self-consistent interior field and spacetime solution adds a measurable correction beyond a matched baseline. Gravitational-wave mode shifts, mass-radius changes, tidal response, ellipticity, and polarization signatures are similarly defined as targets for calculation rather than pre-existing evidence.

Final scientific position. HMGQM v3.0 is a proposed effective framework, not a verified law of nature. Its value will ultimately be decided by three tests: (1) mathematical self-consistency, (2) robustness against microphysical and geometric uncertainties, and (3) the existence or absence of a distinctive falsifiable observational signature.

33 Thermodynamic Ensemble and Finite-Temperature Extension

33.1 Why the ensemble must be stated

The local thermodynamic quantities depend on the choice of independent variables. In the cold-star limit it is convenient to work at $T = 0$ and specify chemical potentials, baryon density, and magnetic field. For proto-neutron stars, merger remnants, or hot crusts, temperature is not negligible and the grand potential must be treated as a function of (T, μ_i, B). The HMGQM architecture does not depend on the zero-temperature simplification, but the closure must state the ensemble explicitly.

At finite temperature,

$$\mathrm{d}\Omega = -s\,\mathrm{d}T - \sum_i n_i\,\mathrm{d}\mu_i - M\,\mathrm{d}B, \tag{75}$$

and the energy density becomes

$$\varepsilon = \Omega + Ts + \sum_i \mu_i n_i + \frac{B^2}{2}. \tag{76}$$

The pressure identities retain the same field-stress structure. This extension is important because a model intended for a merger remnant cannot be justified by a cold EOS alone.

33.2 Entropy and consistency

The entropy density is

$$s = -\left(\frac{\partial\Omega}{\partial T}\right)_{\mu_i,B}. \tag{77}$$

All thermodynamic derivatives should be obtained from one numerical representation of Ω. Mixing a zero-temperature density table with a separately approximated magnetization is discouraged because it can violate integrability.

34 Local Relativistic Frames and the Meaning of Magnetic Field

34.1 Comoving observer

The scalar magnetic field used in the EOS is the field measured in the local fluid rest frame. Given u^μ, the electric and magnetic fields can be defined covariantly from $F_{\mu\nu}$. The local magnetic magnitude is

not generally equal to a coordinate component such as $F_{r\theta}$.

34.2 Magnetic four-direction

Let b^μ denote the unit spacelike direction of the magnetic field in the comoving frame. Then

$$u^\mu u_\mu = -1, \qquad b^\mu u_\mu = 0, \qquad b^\mu b_\mu = 1. \tag{78}$$

These relations make the principal-stress decomposition coordinate-independent.

34.3 Why this matters numerically

A code may evolve electromagnetic potentials in coordinates while the EOS expects a local invariant field strength. The conversion between the coordinate representation and the comoving B must be performed at every grid point. This avoids mixing coordinate artifacts with physical anisotropy.

35 Magnetization, Ensembles, and Thermodynamic Bookkeeping

35.1 Magnetization is a derivative

The magnetization is not an independent free parameter in the minimal theory. It is the derivative

$$M = -\left(\frac{\partial \Omega}{\partial B}\right)_{T,\mu_i} \tag{79}$$

of the same potential that defines the matter densities. This derivative includes all explicitly B-dependent matter terms and the renormalized QED contribution included in Ω.

35.2 Choice of external versus self-consistent field

Thermodynamics formulated at fixed external B is not identical to a fully self-consistent magnetostatic stellar problem. In the latter, currents and the field are solved together. HMGQM therefore uses the local thermodynamic potential for the microscopic closure but does not treat B as a globally prescribed scalar field.

35.3 Avoiding double counting

If $\Omega_{\rm HE}$ is already part of the thermodynamic potential, its derivative contributes to M and its stress contribution enters $T_{\mu\nu}$. A second manually inserted "vacuum pressure" would duplicate that physics. The action and thermodynamic potential are therefore the single sources of truth.

36 Electromagnetic Current Closure

36.1 Why Maxwell's equation needs a current

The nonlinear equation

$$\nabla_\mu H^{\mu\nu} = J^\nu \tag{80}$$

requires a physically specified current. In an equilibrium star, the current is not arbitrary: it must be compatible with the matter motion, charge distribution, stationarity, and chosen field topology.

36.2 Current function

For a stationary axisymmetric model, a current function can be introduced as part of the equilibrium ansatz. Different choices generate different field configurations even if the central field amplitude is

the same. Therefore a complete HMGQM parameter set must include the current prescription or an equivalent field-topology specification.

36.3 Gauge freedom

The electromagnetic potentials are defined only up to $A_\mu \to A_\mu + \nabla_\mu \Lambda$. Observable fields depend on $F_{\mu\nu}$, not the gauge choice. Numerical tests must verify gauge-independent physical outputs.

37 Surface Conditions and Exterior Matching

37.1 Interior boundary

The stellar surface is not necessarily a surface of constant coordinate radius. It is defined by the thermodynamic condition appropriate to the chosen EOS, typically where the matter density falls to the crust or atmosphere matching density.

37.2 Metric matching

The first and second fundamental forms must be matched consistently between the interior and exterior spacetime. The precise implementation depends on whether the configuration is static, rotating, magnetized, or includes a magnetosphere.

37.3 Electromagnetic matching

The normal and tangential electromagnetic fields obey the appropriate Maxwell boundary conditions. Surface currents, if present, must be explicitly represented. A model that simply truncates the interior field at an arbitrary radius can generate unphysical surface stresses.

37.4 Exterior radiation

For polarization predictions, the exterior is part of the transport problem. A vacuum approximation is acceptable only when its domain is justified; otherwise the magnetospheric plasma and its current system must be modelled.

38 Rotation, Frame Dragging, and Misaligned Axes

38.1 Stationary versus static

A nonrotating magnetic star can be represented by a static axisymmetric metric. Once rotation is included, the shift term $\omega(r, \theta)$ appears and frame dragging becomes part of the geometry. The two problems should not be mixed.

38.2 Alignment

An aligned, stationary magnetic field is the simplest equilibrium configuration. A misaligned magnetic and rotation axis generally implies time dependence in the inertial frame. Precession can therefore not be inserted into a strictly stationary model without an additional approximation.

38.3 Implication for the original diagram

The precession shown in the original conceptual figure is retained only as an illustration of a possible rotating configuration. It is not a dynamical prediction of the stationary HMGQM equations. A true precession calculation requires time-dependent relativistic magnetohydrodynamics or a controlled perturbative treatment.

39 Global Energetics and Virial Checks

39.1 Energy inventory

For each equilibrium solution, record the gravitational mass, baryon rest mass, internal energy, rotational kinetic energy, magnetic energy, and QED correction. The sum of these contributions should be compatible with the global ADM/Komar mass relation used by the numerical formulation.

39.2 Virial logic

Virial identities provide integral consistency checks linking gravity, pressure, rotation, and magnetic stress. They do not create a new EOS, but they can expose a solution that has apparently converged pointwise while violating global force balance.

39.3 Magnetic-to-gravitational ratio

The dimensionless ratio

$$\mathscr{E}_B = \frac{E_B}{|W|} \tag{81}$$

with W a gravitational binding-energy measure, provides an immediately interpretable indicator of magnetic back-reaction. It is complementary to the local diagnostic β_B.

40 Dimensionless Scaling and Unit Conversion

Quantity	Natural-unit definition / practical role
B_{QED}	m_e^2/e, the electron QED critical field.
χ_B	B/B_{QED}, field-strength diagnostic.
β_B	$B^2/(2P_m)$, local magnetic-pressure fraction.
$\nu_{\max}$	Highest occupied Landau level; quantization diagnostic.
χ_R	$(\lambda_e/L_{\mathrm{curv}})^2$, local-curvature diagnostic.
κ_{loop}	$\alpha \ln\!\big(1 + B/B_{\mathrm{QED}}\big)$, one-loop control indicator.
Γ_B	$\lambda_e\lvert\nabla B\rvert/B$, constant-field validity indicator.

Table 3: Dimensionless diagnostics used by the complete HMGQM framework.

All code and published plots should state whether B is expressed in tesla, gauss, or natural units. Conversion errors can otherwise imitate changes in physical regimes by many orders of magnitude.

41 Parameter-Space Strategy

41.1 Primary scan variables

A numerical survey should vary central density, central magnetic field, field topology, rotation rate, temperature, and microphysical model. Rather than scanning each quantity independently without physical control, sample along equilibrium sequences where global constraints are maintained.

41.2 Regime boundaries

The code should mark surfaces where $\nu_{\max,i}$ changes integer value, where β_B crosses a chosen dynamical threshold, and where the one-loop control indicators become marginal. These boundaries are numerical diagnostics, not assumed phase transitions.

41.3 Uncertainty bands

For each observable O, report

$$O = O(\Theta), \qquad \sigma_O^2 \simeq \sum_a \left(\frac{\partial O}{\partial \theta_a}\right)^2 \sigma_a^2 \tag{82}$$

when a linear approximation is adequate, or use full Monte Carlo propagation when it is not.

42 Solver Architecture and Pseudocode

A practical implementation can be organized as follows:

1. initialize a microphysical EOS and the thermodynamic potential Ω;
2. solve charge neutrality and beta equilibrium for the composition;
3. evaluate n_i, M, ε, $P_\parallel$, and $P_\perp$;
4. construct $T_{\mu\nu}$ and the constitutive tensor $H^{\mu\nu}$;
5. solve the stationary axisymmetric Einstein-Maxwell equations;
6. update the local magnetic field and matter variables;
7. iterate to convergence;
8. validate residuals and limiting cases;
9. compute global observables;
10. pass the equilibrium to perturbation and radiative-transfer modules.

A solver should stop and report failure when the validity criteria are violated or when conservation residuals fail to converge, rather than forcing convergence through numerical damping.

43 Error Control and Numerical Convergence

43.1 Resolution study

Repeat the same model at increasing spatial resolution. An observable O_N should approach a converged value with the expected numerical order. Report both the estimated truncation error and the change between the final two resolutions.

43.2 Integral versus differential checks

Global quantities such as ADM mass and baryon mass should be computed using at least two independent numerical representations where feasible. Agreement provides a strong cross-check on the discretization.

43.3 EOS interpolation

Interpolation errors in the thermodynamic potential can be more damaging than the formal PDE discretization error because derivatives of an interpolated table directly enter the stress tensor. Monotonicity and thermodynamic integrability should therefore be checked before the table is coupled to GR.

44 Construction of an Observational Likelihood

44.1 Data vector

Let D collect measured masses, radii, polarization spectra, timing parameters, tidal information, or gravitational-wave data. The model maps parameters Θ to predicted observables $O(\Theta)$.

44.2 Likelihood

For approximately Gaussian measurement errors,

$$\ln \mathscr{L} = -\frac{1}{2}[D - O(\Theta)]^T C^{-1}[D - O(\Theta)] + \text{const.} \tag{83}$$

For non-Gaussian data, use the published likelihood or an appropriate hierarchical model.

44.3 Baseline comparison

The central quantity is not merely $\mathscr{L}_H$, but the evidence ratio between the HMGQM model and a matched baseline. The models should share the same priors over nuisance parameters wherever scientifically justified.

45 Polarization-Transport Test Design

The electromagnetic test should proceed in layers. First solve the surface emission model. Next propagate the polarization modes through the curved spacetime and magnetized exterior. Finally compare the phase- and energy-dependent polarization observables against the baseline model.

The 2026 magnetar observations demonstrate why this pipeline is relevant: strong polarization is observable, but the measured signal reflects the combined effects of surface emission geometry, QED birefringence, and magnetospheric transport. HMGQM therefore gains empirical content only by changing the complete forward model, not by asserting that birefringence exists.

A useful report includes the posterior distributions of the field geometry, viewing angle, surface temperature distribution, polarization degree, polarization angle, and any finite-field QED correction. A single best-fit curve is not enough to establish discrimination.

46 Gravitational-Wave Test Design

46.1 Equilibrium first

A waveform or mode spectrum cannot be computed consistently before an equilibrium star is defined. The metric, matter distribution, magnetic field, and stress anisotropy are all inputs to the perturbation problem.

46.2 Mode identification

Label modes by their dominant character and track them continuously along an equilibrium sequence. Avoid assigning mode identities solely from frequency ordering when magnetic coupling causes avoided crossings.

46.3 Matched waveform analysis

For a merger or continuous-wave calculation, generate a baseline waveform and an HMGQM waveform at the same macroscopic parameters. The discriminating residual is

$$\Delta h(t) = h_H(t) - h_0(t). \tag{84}$$

Whether this residual is measurable requires comparison to detector sensitivity and astrophysical uncertainty, not visual inspection alone.

47 Phase-Transition Search Protocol

47.1 Thermodynamic scan

Compute $\Omega(\rho, B, T)$ on a dense grid and inspect derivatives with respect to the control variables. Search for discontinuities, divergences, or multiple locally stable branches separated by a free-energy crossing.

47.2 Coexistence condition

A first-order transition requires equality of the relevant thermodynamic potentials between competing phases. One then computes the coexistence curve in (ρ, B, T) space and the associated discontinuities in density, entropy, or magnetization.

47.3 If no transition exists

If the scan reveals only continuous crossovers, the correct conclusion is that H-Matter is a useful regime classification. The absence of a phase transition does not invalidate the predicted anisotropic stresses or observational consequences.

48 Null Tests and Counterfactual Models

A robust HMGQM study should construct several null models. The first removes the QED vacuum correction but retains Landau quantization. The second retains QED photon propagation but suppresses its contribution to the interior stress-energy. The third removes Landau quantization while retaining classical magnetic stress. These counterfactuals isolate which physical ingredient causes an observable difference.

This decomposition is essential because a measured effect could otherwise be attributed to the wrong mechanism. For example, strong polarization might arise from ordinary magnetospheric geometry even when finite-field QED corrections are small. A controlled hierarchy of null models identifies the incremental information carried by the HMGQM coupling.

49 Reproducibility Standard

A publication-quality HMGQM calculation should publish the exact microscopic EOS or a machine-readable reference, the numerical grid and convergence settings, the magnetic topology prescription, all control parameters, and enough information to reconstruct the dimensionless diagnostics. The code

need not be public to establish reproducibility, but the mathematical and numerical inputs must be specified sufficiently that an independent group could implement them.

Every figure should distinguish between measured data, baseline-model curves, HMGQM curves, and purely schematic illustrations. Any quantity described as a "prediction" should have a mathematical route from the master system to that quantity.

50 Publication and Peer-Review Checklist

Before presenting HMGQM as a research result, verify:

1. every equation has a stated unit convention;
2. the microscopic EOS is explicitly identified;
3. all field and matter contributions to $T_{\mu\nu}$ come from one bookkeeping system;
4. the strong-field QED expression is used within its domain;
5. the global geometry matches the symmetry assumptions;
6. equilibrium residuals and convergence tests are reported;
7. stability claims are based on a justified perturbation analysis;
8. observational claims use only quantities actually measured by the cited data set;
9. baseline and HMGQM models are compared at matched parameters;
10. uncertainty propagation is included;
11. the phase label is not promoted to a fundamental substance without thermodynamic evidence;
12. the numerical and observational failure modes are stated explicitly.

51 Scientific Status of the Complete Formulation

HMGQM v3.0 should be understood as a disciplined research program rather than a finished replacement for established physics. The framework is strong enough to define a coupled calculation, but it remains open at the microscopic closure, global numerical solution, perturbation spectrum, and observational discrimination stages.

The distinction matters. A theory becomes scientifically persuasive not when its equations look sophisticated, but when it produces stable, controlled solutions and predictions that survive alternative microphysics, numerical tests, and independent observations. HMGQM v3.0 is structured around precisely those requirements.

Status statement. The framework is mathematically specified at the effective-action level but is not yet an empirically established theory of a new state of matter. The remaining work is computational and observational, with the phase question deliberately left open.

A Appendix A – Detailed Thermodynamic Derivation

A.1 Grand-canonical relations

For a homogeneous local fluid element, the grand potential density obeys

$$\mathrm{d}\Omega = -s\,\mathrm{d}T - \sum_i n_i\,\mathrm{d}\mu_i - M\,\mathrm{d}B, \tag{85}$$

so that

$$s = -\frac{\partial\Omega}{\partial T}, \qquad n_i = -\frac{\partial\Omega}{\partial\mu_i}, \qquad M = -\frac{\partial\Omega}{\partial B}. \tag{86}$$

At zero temperature, the first term disappears.

A.2 Differential structure and consistency

A numerical implementation should verify the equality of mixed derivatives, for example

$$\frac{\partial n_i}{\partial B} = \frac{\partial M}{\partial\mu_i}, \tag{87}$$

when the derivatives are evaluated within the same thermodynamic convention. Failure of such relations signals an inconsistency in the EOS implementation.

A.3 Pressure derivation

The longitudinal pressure follows from the grand potential plus the longitudinal Maxwell stress. The transverse pressure acquires the magnetization term because changing the field at fixed thermodynamic state changes the free energy. Hence

$$P_\parallel = -\Omega - \frac{B^2}{2}, \tag{88}$$

$$P_\perp = -\Omega - MB + \frac{B^2}{2}. \tag{89}$$

Their difference is

$$P_\perp - P_\parallel = B^2 - MB. \tag{90}$$

This compact relation is one of the central diagnostic equations of the framework.

B Appendix B – Landau-Level Summation Details

B.1 Density of states

The transverse motion of a charged fermion in a uniform magnetic field has degeneracy proportional to $|qB|/(2\pi)$ per unit area, while the longitudinal momentum remains continuous. Including the spin degeneracy gives the zero-temperature number density in equation (28).

B.2 Lowest-level degeneracy

The lowest level has a single spin state for a minimally coupled Dirac fermion, captured by

$$g_\nu = 2 - \delta_{\nu 0}. \tag{91}$$

This detail matters because an incorrect degeneracy factor produces a systematic density error precisely in the regime where the HMGQM proposal is most sensitive.

B.3 Many-level limit

When $\nu_{\max} \gg 1$, Euler-Maclaurin or numerical summation should recover the continuum integral. This limit is useful both analytically and computationally. It is also a conceptual guardrail: HMGQM must not create a spurious magnetic effect simply because the code is summing discrete levels when the physical system has many occupied levels.

B.4 Anomalous magnetic moments

If anomalous magnetic moments are included, the spectrum is modified and the simple $m_{i\nu}^2 = m_i^{*2} + 2\nu|q_i|B$ expression must be replaced by the chosen effective dispersion relation. The framework remains unchanged, but the model must report the exact Hamiltonian or effective Lagrangian used for that sector.

C Appendix C – Curvature and Effective-Theory Control

C.1 Local curvature scale

Let L_{curv} denote the shortest relevant curvature radius extracted from scalar curvature invariants or eigenvalues of the Riemann tensor. The dimensionless ratio

$$\chi_R = \left(\frac{\lambda_e}{L_{\text{curv}}}\right)^2 \tag{92}$$

measures whether electron-scale local flatness is a sensible zeroth-order approximation.

C.2 One-loop QED control

The one-loop strong-field action can contain logarithmic growth. The diagnostic

$$\kappa_{\text{loop}} = \alpha \ln\!\left(1 + B/B_{\text{QED}}\right) \tag{93}$$

can be used to flag regions where higher-loop terms may become important. This is not a theorem that one-loop QED fails exactly at a particular value; it is a control parameter informing the truncation error.

C.3 Field gradients

The local constant-field approximation becomes questionable if the magnetic field changes significantly across a Compton wavelength. A code should therefore monitor a dimensionless gradient measure, for example

$$\Gamma_B = \lambda_e \frac{|\nabla B|}{B}. \tag{94}$$

The minimal treatment requires $\Gamma_B \ll 1$ throughout the domain where the Heisenberg-Euler sector is used.

D Appendix D – Global Compactness and Horizon Avoidance

D.1 Spherical reference criterion

For a spherical star, a useful reference compactness is

$$C = \frac{GM}{Rc^2}. \tag{95}$$

A Schwarzschild horizon would occur at $C = 1/2$ in the spherical vacuum matching problem. HMGQM v3.0 does not simply impose $2GM/(Rc^2) < 1$ on a deformed star and call the problem solved; instead, horizon avoidance is assessed from the actual stationary spacetime. The spherical condition is used only as a reference scale.

D.2 Axisymmetric caution

For a rotating or strongly magnetized object, the geometry is not characterized by one radius. Equatorial and polar circumferential radii differ, and the horizon condition is a property of the spacetime foliation. Numerical solutions must therefore demonstrate regularity and absence of a trapped-surface/horizon structure within the intended domain.

D.3 Why this matters

The original draft repeatedly used the phrase "below event-horizon formation" without a mathematical criterion. The complete framework replaces that phrase with explicit compactness diagnostics and a requirement that the actual axisymmetric solution remain in the regular, horizon-free branch being studied.

E Appendix E – Glossary of Core Terms

Term	Meaning in HMGQM v3.0
HMGQM	Hyper-Dense Magneto-Gravitational Quantum Matter framework.
H-Matter	Operational label for the high-field/high-density regime defined in this document; a genuine phase requires further thermodynamic evidence.
H-regime	The dimensionless intersection of strong field, strong quantization, and non-negligible magnetic/QED back-reaction within the theory's validity domain.
B_{QED}	Electron QED critical field m_e^2/e.
Landau quantization	Discretization of charged-particle transverse motion in a magnetic field.
Heisenberg-Euler sector	One-loop QED effective action describing nonlinear vacuum response to electromagnetic fields.
Magnetization M	Thermodynamic response $-\partial\Omega/\partial B$.
$P_{\parallel}$	Total principal stress parallel to the local magnetic field.
$P_{\perp}$	Total principal stress transverse to the local magnetic field.
Einstein-Maxwell system	Coupled general-relativistic gravity and nonlinear electromagnetic equations.
AMM	Anomalous magnetic moment correction.
EOS	Equation of state or microscopic thermodynamic closure.
GRMHD	General-relativistic magnetohydrodynamics.
QED	Quantum electrodynamics.
NICER	Neutron Star Interior Composition Explorer X-ray telescope.
IXPE	Imaging X-ray Polarimetry Explorer.
GW170817	First observed binary-neutron-star gravitational-wave event with an electromagnetic counterpart.

GW190814	Compact-binary event with a $\sim 2.6M_\odot$ secondary and a $\sim 23M_\odot$ primary black hole candidate.

F Appendix F – Recommended Computational Experiments

F.1 Experiment 1: one-species Landau ladder

Fix μ^* and sweep B. Record $\nu_{\max}$, n, Ω, M, $P_\parallel$, and $P_\perp$. Demonstrate the approach to the LLL and the recovery of the continuum limit.

F.2 Experiment 2: QED finite-field check

Evaluate the proper-time Heisenberg-Euler integral and compare it to its weak-field expansion over a broad range of B/B_{QED}. Plot the relative error and identify where the expansion becomes unreliable.

F.3 Experiment 3: spherical toy star

Use an anisotropic spherical model only as an educational intermediate step. Confirm that the anisotropy term disappears as $B \to 0$ and that the model converges toward ordinary TOV.

F.4 Experiment 4: full axisymmetric star

Turn on a self-consistent poloidal field and compare equatorial and polar radii. Monitor residuals of all field equations. Repeat with a mixed poloidal-toroidal topology.

F.5 Experiment 5: matched observables

Choose a fixed mass and surface field. Run baseline and HMGQM models with the same EOS and rotation. Compare R, I, ϵ_B, Λ, polarization, and mode frequencies.

F.6 Experiment 6: phase search

Compute the thermodynamic potential over density and field. Search for nonanalyticities in first and second derivatives. If none appear, document the absence of a phase transition and retain H-Matter as a regime label.

G Appendix G – Reference Notes

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H Appendix H – Final Research Checklist

Requirement	Status in v3.0
Single effective action	☒
Strong-field QED treated at finite field	☒
Landau quantization included explicitly	☒
Species-specific treatment	☒
Thermodynamically consistent stresses	☒
No double counting of magnetic/vacuum stress	☒
Axisymmetric global equilibrium	☒
Explicit magnetic topology requirement	☒
Stability hierarchy	☒
Causality checks	☒
Known limiting cases	☒
Observational discipline	☒
Falsifiability criteria	☒
Phase claim requires thermodynamic evidence	☒
Quantum gravity kept outside the minimal theory	☒
Numerical implementation pathway	☒
Author note and hyperlink-only contacts	☒

Final statement in one line: HMGQM is a proposed, self-consistent effective framework for the coupled high-density, ultra-magnetized, QED-corrected relativistic regime of compact matter; its scientific status depends on whether the resulting solutions are mathematically sound, physically controlled, and observationally distinguishable from established magnetized-matter models.

End of complete formulation – HMGQM Theory v3.0 – October 2026